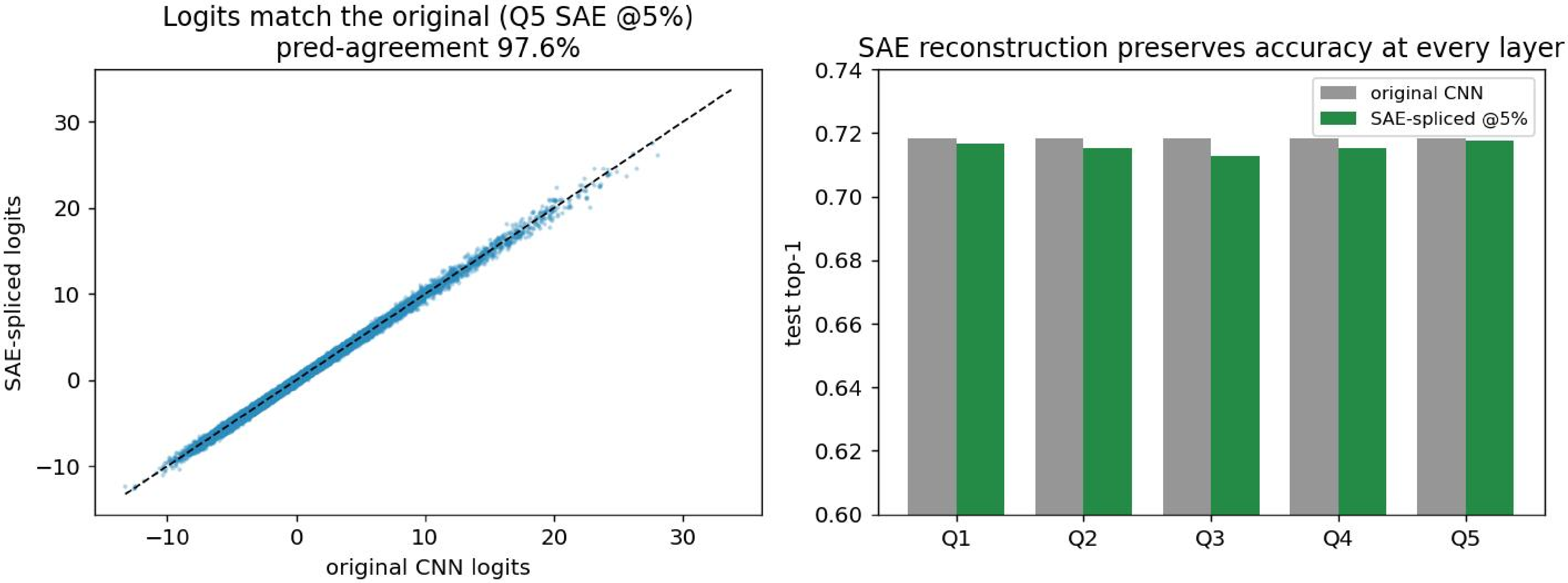
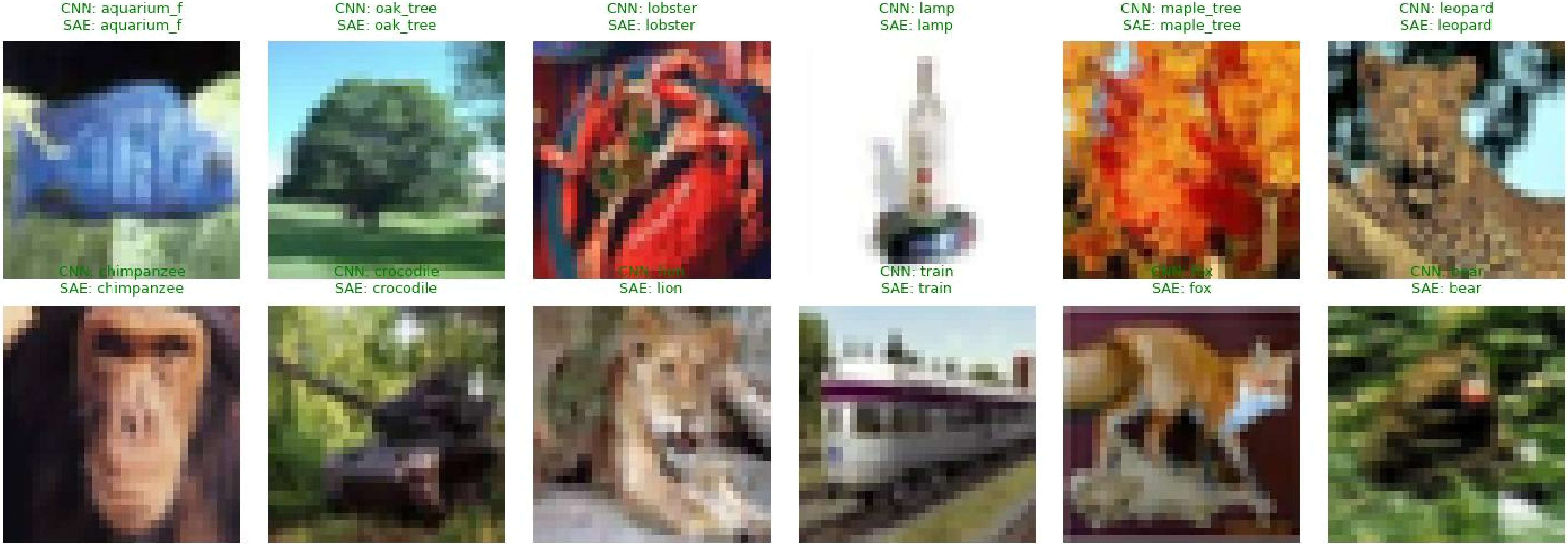
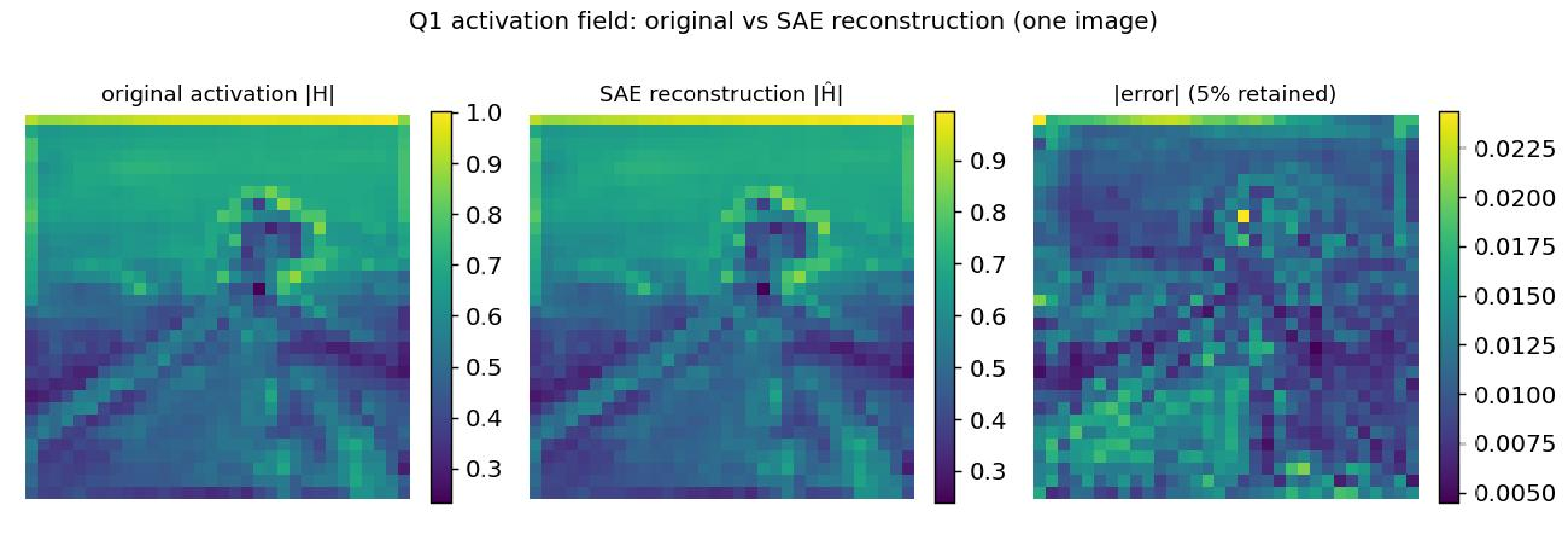
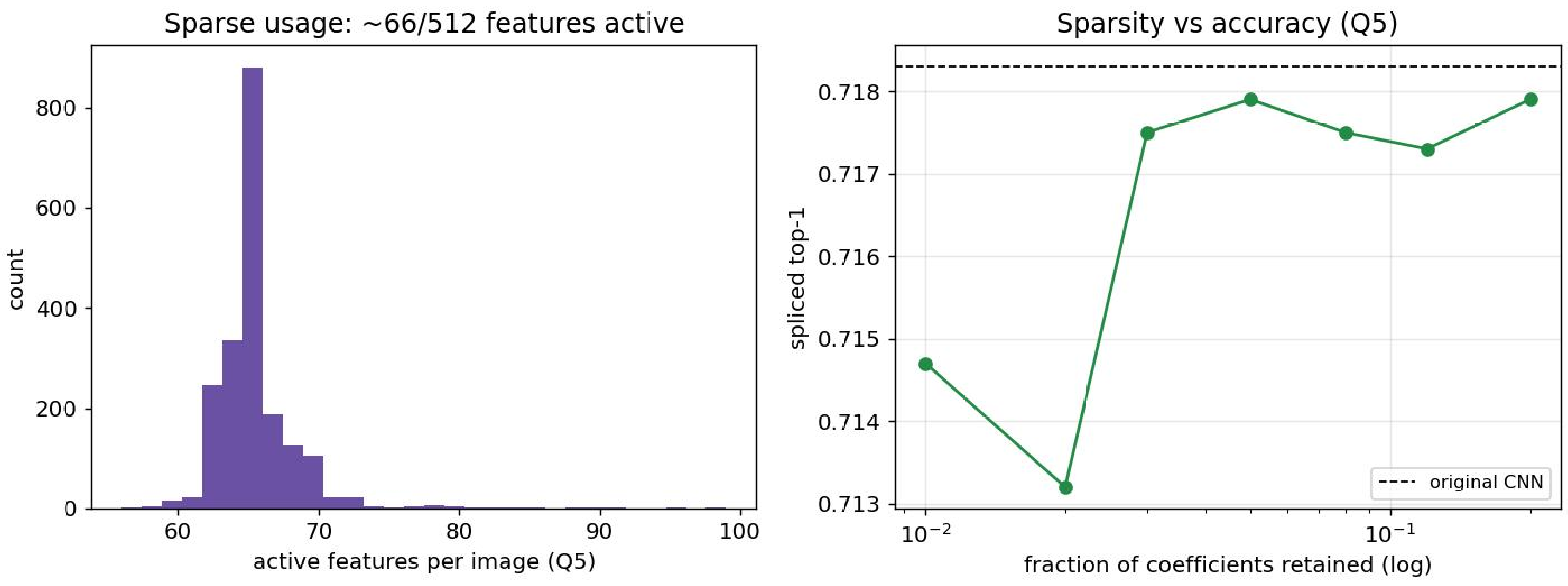


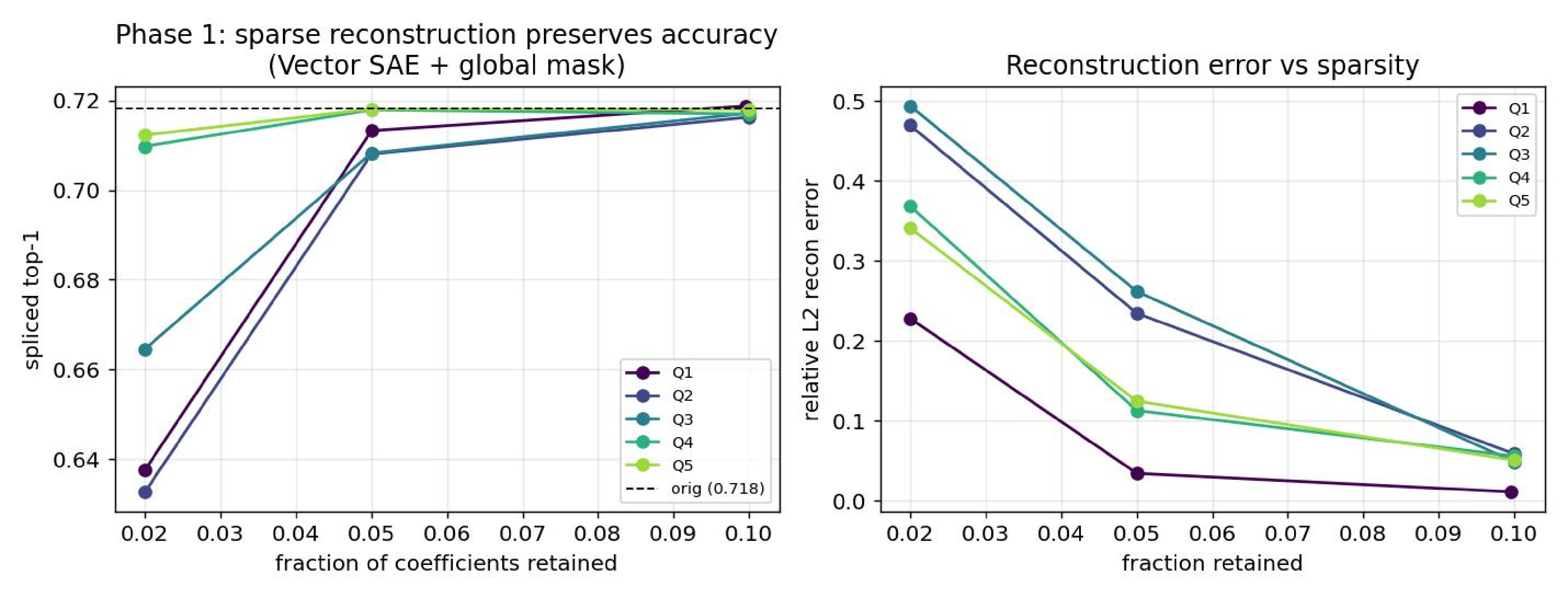
Logits match the original (Q5 SAE @5%)
pred-agreement 97.6%
SAE-spliced logits
original CNN logits
SAE reconstruction preserves accuracy at every layer
original CNN
SAE-spliced @5%
test top-1
Q1
Q2
Q3
Q4
Q5

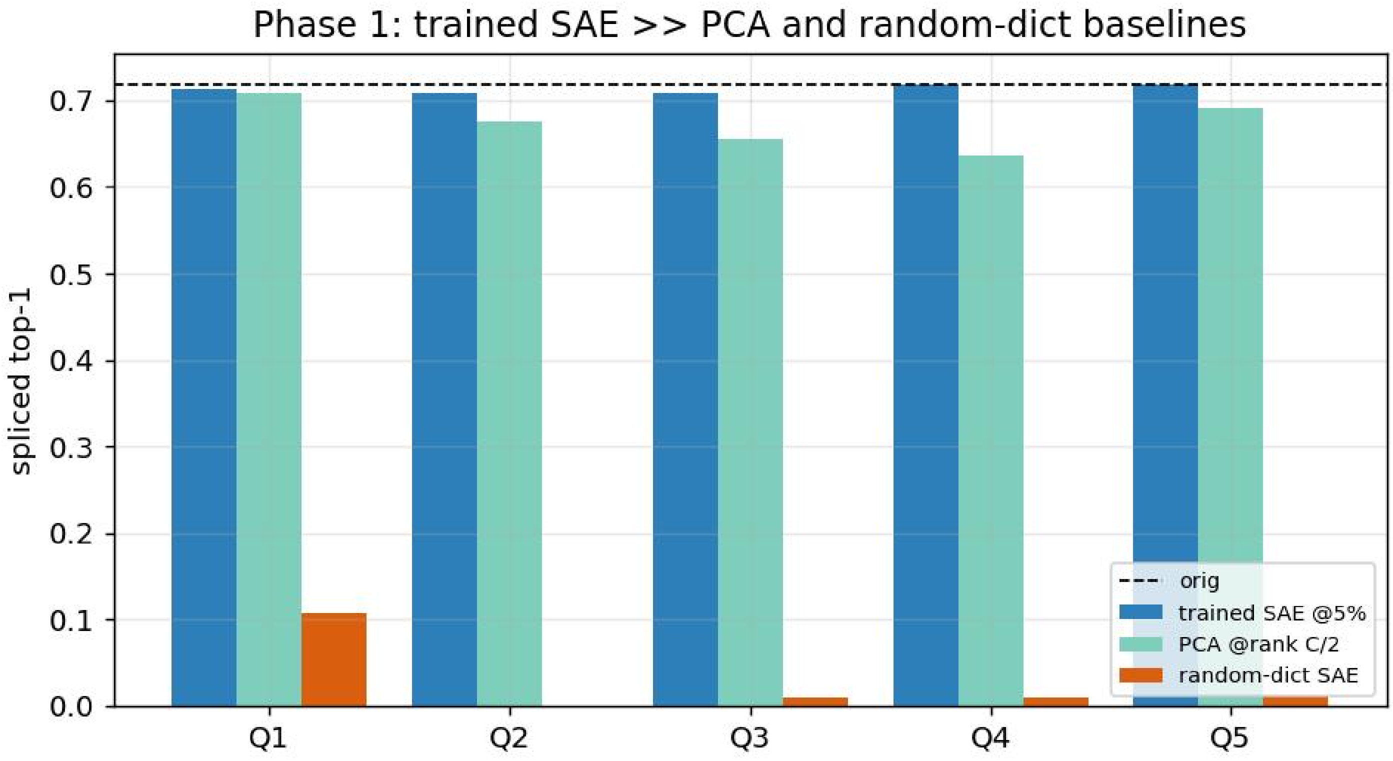
Original CNN prediction vs SAE-spliced prediction (Q5 @5% retained)

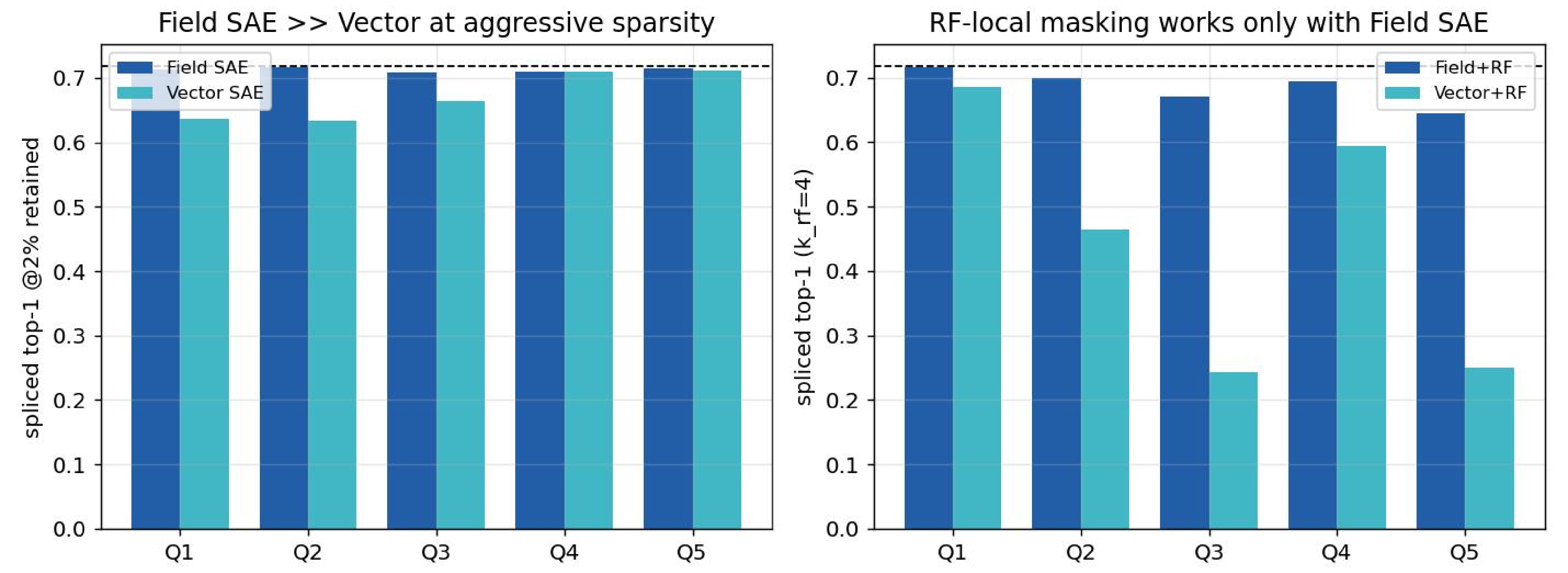



Q1 activation field: original vs SAE reconstruction (one image)
original activation |H|
SAE reconstruction |Ĥ|
|error| (5% retained)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.0225
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050

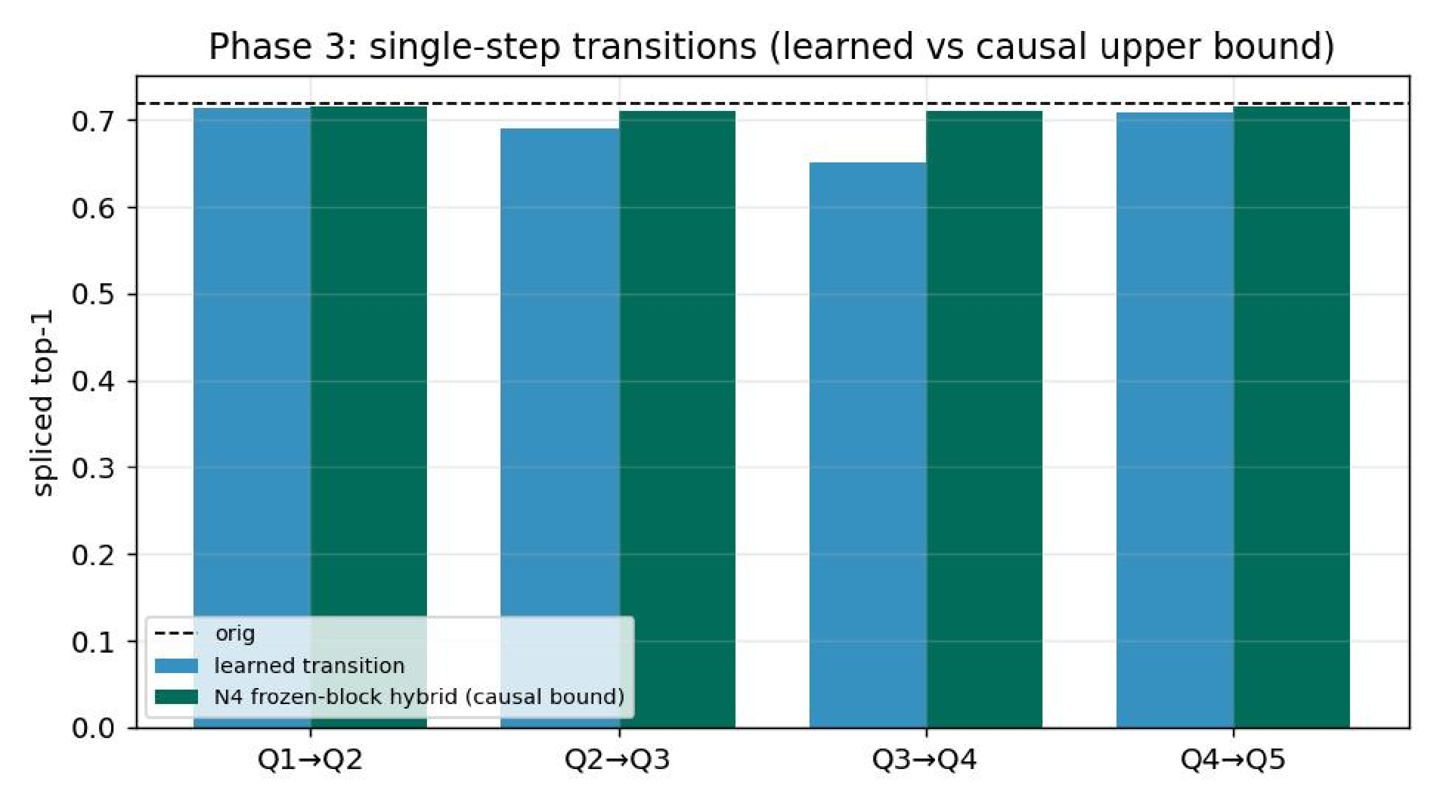

Sparse usage: ~66/512 features active
count
800
600
400
200
0
60
70
80
90
100
active features per image (Q5)
Sparsity vs accuracy (Q5)
spliced top-1
0.718
0.717
0.716
0.715
0.714
0.713
10^{-2}
10^{-1}
fraction of coefficients retained (log)
original CNN


Phase 1: sparse reconstruction preserves accuracy
(Vector SAE + global mask)
spliced top-1
fraction of coefficients retained
0.72
0.70
0.68
0.66
0.64
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.10
Q1
Q2
Q3
Q4
Q5
orig (0.718)
Reconstruction error vs sparsity
relative L2 recon error
fraction retained
0.5
0.4
0.3
0.2
0.1
0.0
Q1
Q2
Q3
Q4
Q5

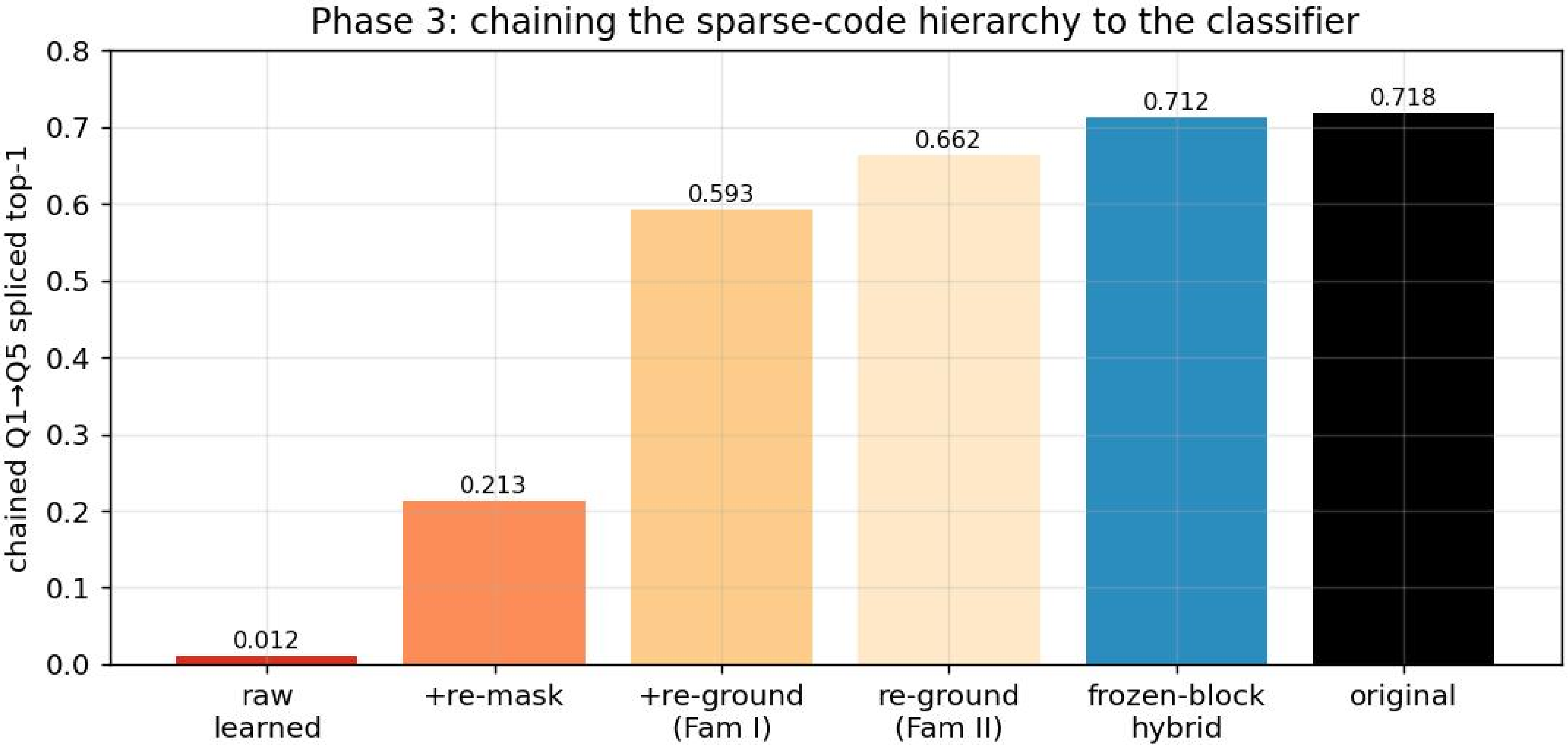

Phase 1: trained SAE >> PCA and random-dict baselines
spliced top-1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Q1
Q2
Q3
Q4
Q5
orig
trained SAE @5%
PCA @rank C/2
random-dict SAE



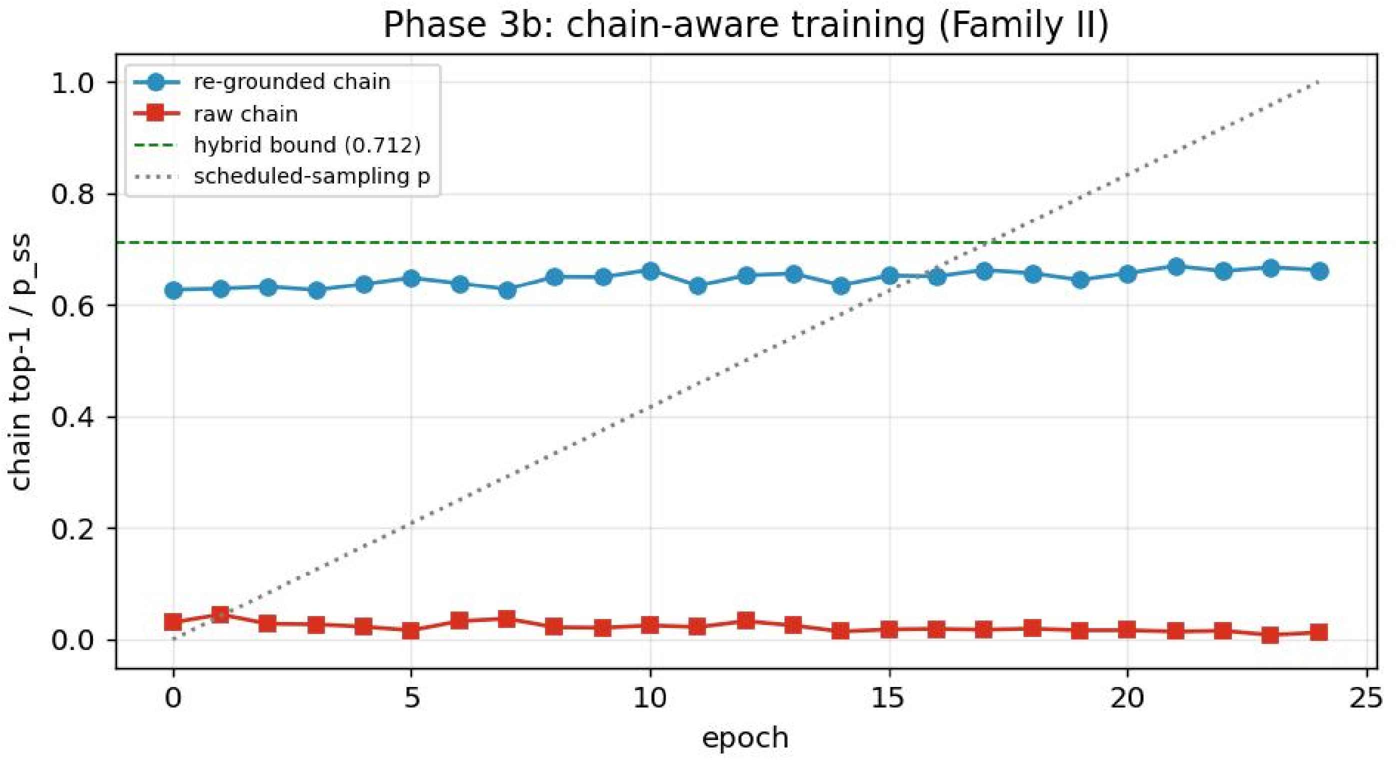
Field SAE >> Vector at aggressive sparsity
Field SAE
Vector SAE
spliced top-1 @2% retained
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Q1
Q2
Q3
Q4
Q5
RF-local masking works only with Field SAE
Field+RF
Vector+RF
spliced top-1 (k_rf=4)



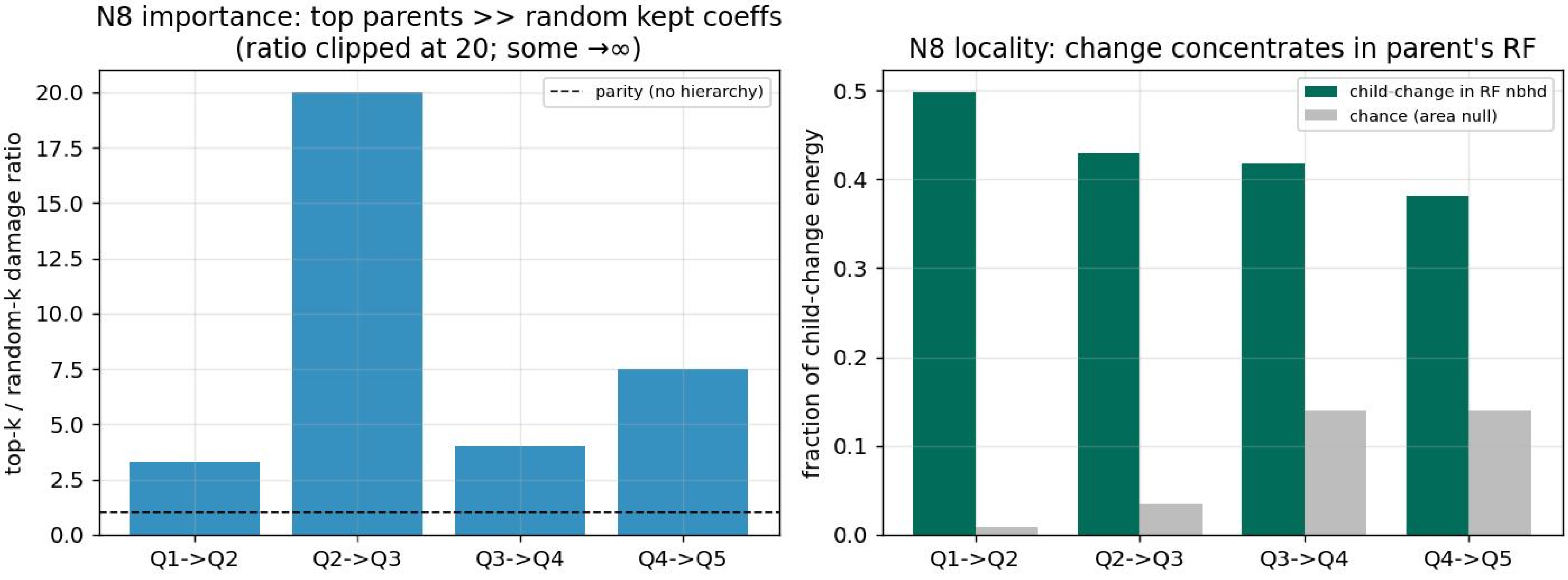
Phase 3: single-step transitions (learned vs causal upper bound)
spliced top-1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Q1→Q2
Q2→Q3
Q3→Q4
Q4→Q5
orig
learned transition
N4 frozen-block hybrid (causal bound)

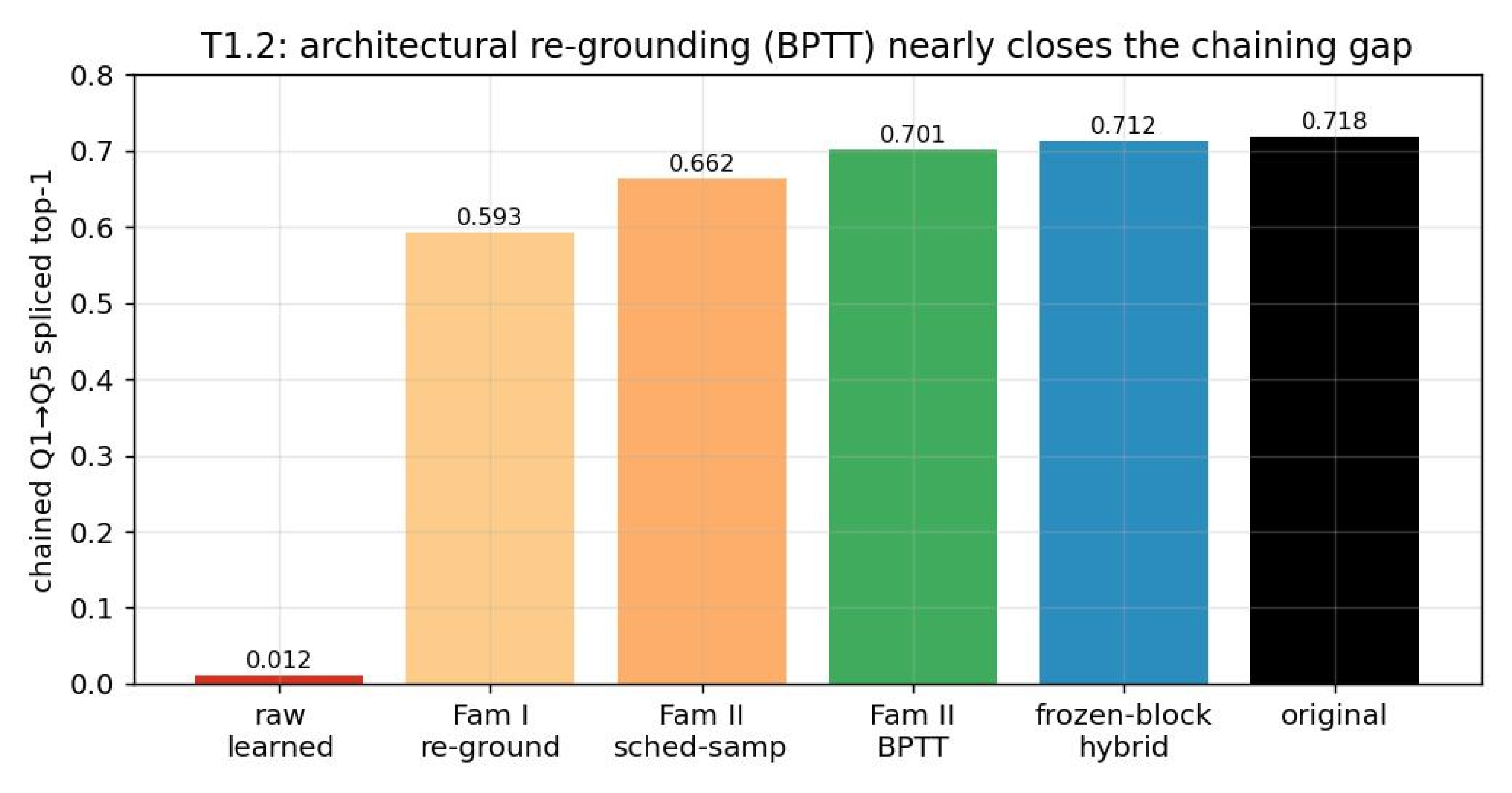


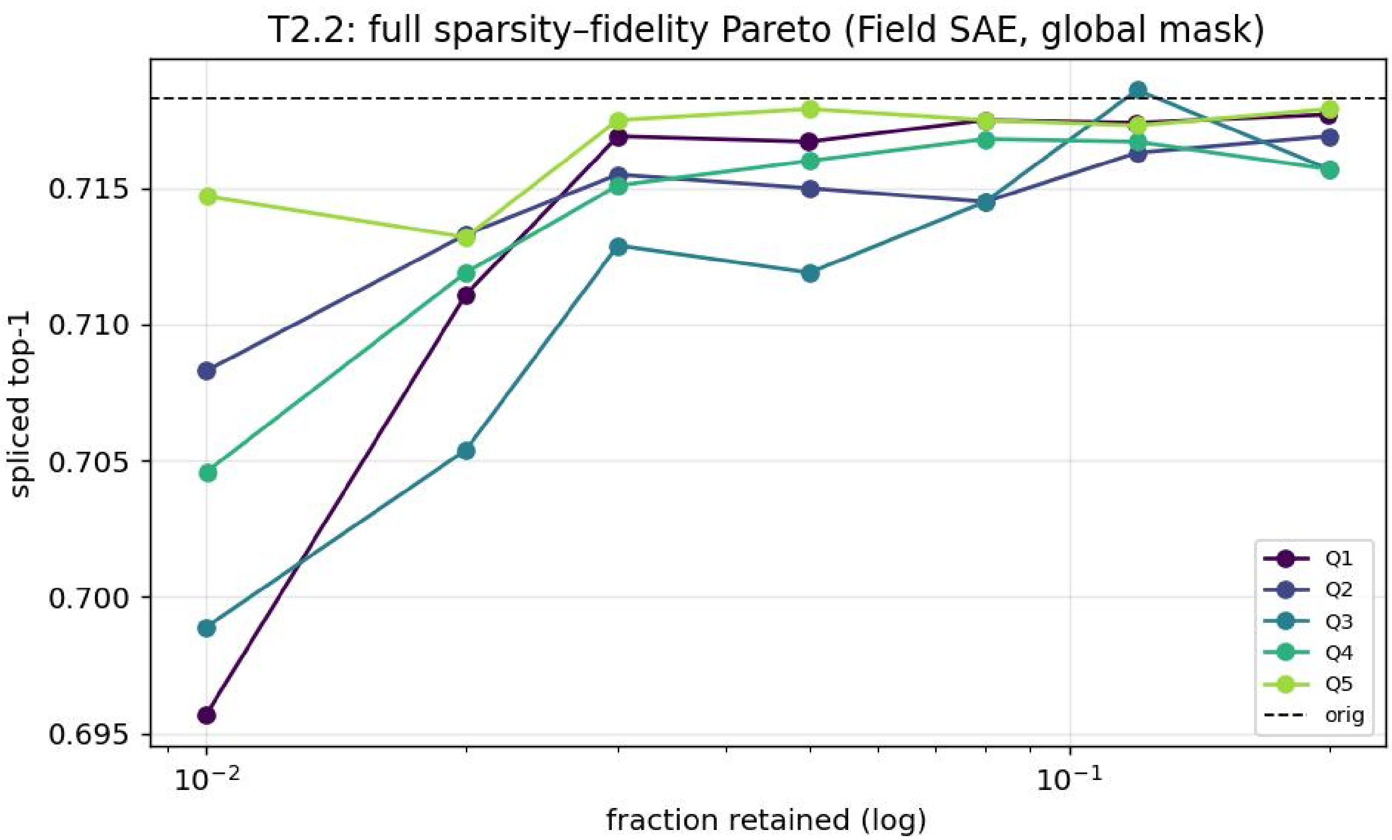
Phase 3: chaining the sparse-code hierarchy to the classifier
chained Q1→Q5 spliced top-1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.012
0.213
0.593
0.662
0.712
0.718
raw learned
+re-mask
+re-ground (Fam I)
re-ground (Fam II)
frozen-block hybrid
original



Phase 3b: chain-aware training (Family II)
re-grounded chain
raw chain
hybrid bound (0.712)
scheduled-sampling p
chain top-1 / p_ss
epoch
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25


N8 importance: top parents >> random kept coeffs
(ratio clipped at 20; some →∞)
parity (no hierarchy)
top-k / random-k damage ratio
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Q1->Q2
Q2->Q3
Q3->Q4
Q4->Q5
N8 locality: change concentrates in parent's RF
child-change in RF nbhd
chance (area null)
fraction of child-change energy
0.0
0.1
0.2
0.3
0.4
0.5
Q1->Q2
Q2->Q3
Q3->Q4
Q4->Q5


T1.2: architectural re-grounding (BPTT) nearly closes the chaining gap
chained Q1→Q5 spliced top-1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.012
0.593
0.662
0.701
0.712
0.718
raw learned
Fam I re-ground
Fam II sched-samp
Fam II BPTT
frozen-block hybrid
original



T2.2: full sparsity–fidelity Pareto (Field SAE, global mask)
spliced top-1
0.715
0.710
0.705
0.700
0.695
10^{-2}
10^{-1}
fraction retained (log)
Q1
Q2
Q3
Q4
Q5
orig